

PharmacyGPT

AI-Powered Pharmaceutical Assistant

DEPI Graduation Project Presentation

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Project Idea



Problem Statement

Patients often struggle to identify possible illnesses based on their symptoms and choose suitable over-the-counter (OTC) medications. Existing solutions may not reflect the Egyptian pharmaceutical market or provide personalized preliminary guidance.



Proposed Solution

Pharmacy GPT is an AI-powered web application that analyzes a user's age, gender, symptoms, and medical notes to predict possible medical conditions and recommend appropriate OTC medications available in Egypt. The system also provides medicine precautions, home-care advice, and follow-up assistance through an AI chatbot.



Unique Value Proposition



Egyptian Market Focus

Recommendations tailored to the Egyptian pharmaceutical market



AI-Powered with Llama 3.2

Powered by Ollama (Llama 3.2) for intelligent, context-aware responses



Self-Expanding Knowledge Base

automatically learns from new cases and expands the medical dataset dynamically



Interactive AI Chatbot

Provides follow-up answers, dosage guidance, and health advice.



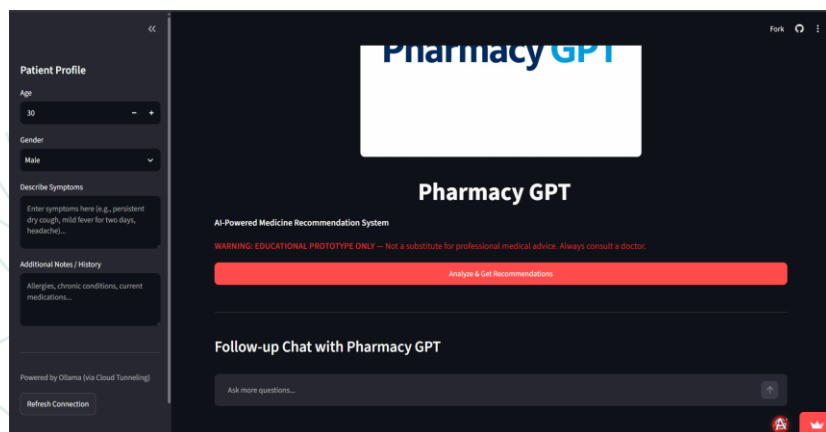
Hybrid Intelligence Approach

Combines structured medical datasets with LLMs for more accurate and reliable recommendations.

Project Wireframe

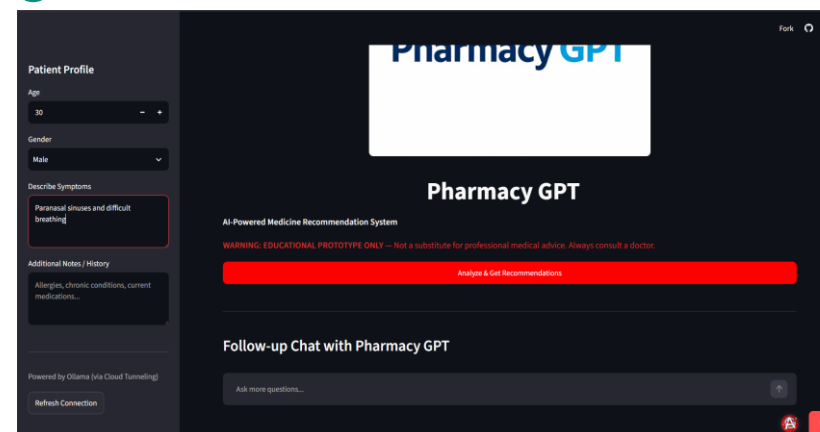
Key User Interfaces (Wireframes)

1 Home Page



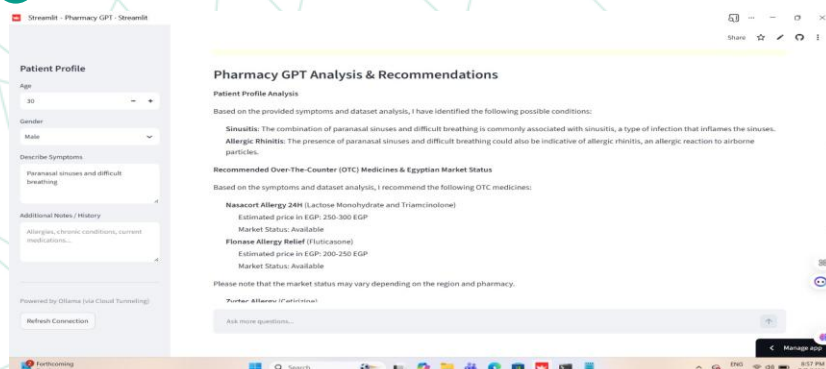
The Home Page wireframe shows a dark-themed interface. On the left is a 'Patient Profile' sidebar with fields for Age (30), Gender (Male), and a 'Describe Symptoms' section. The main area features the 'Pharmacy GPT' logo, a warning about the educational prototype, a red 'Analyze & Get Recommendations' button, and a 'Follow-up Chat with Pharmacy GPT' section at the bottom.

2 Symptom Input Form



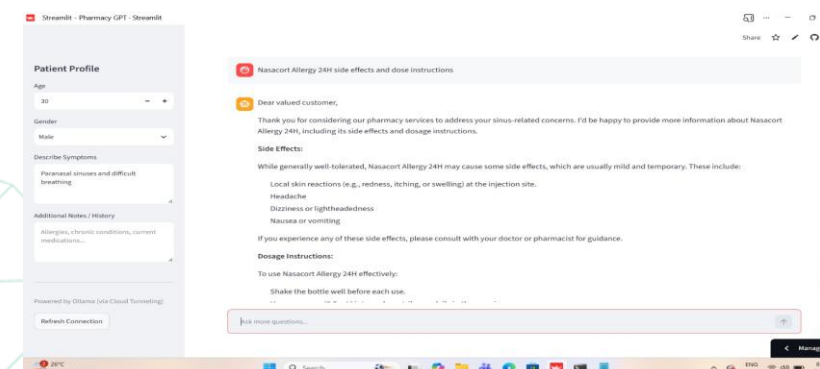
The Symptom Input Form wireframe is identical to the Home Page, but the 'Describe Symptoms' section in the sidebar is highlighted with a red border, indicating the input area for the user's symptoms.

3 Results & Recommendation



The Results & Recommendation wireframe shows the 'Pharmacy GPT Analysis & Recommendations' section. It includes a 'Patient Profile Analysis' summary, a list of recommended OTC medicines (Nasacort Allergy 24H and Flonase Allergy Relief), and their market status. The sidebar remains on the left.

4 AI Chatbot



The AI Chatbot wireframe shows a chat interface with a 'Patient Profile' sidebar. The chat area displays a message from the AI chatbot providing detailed information about Nasacort Allergy 24H, including side effects and dosage instructions.

User Journey



Usability & User Experience



End Users & Features



END USERS (who will use PharmacyGPT?)



General Users

Individuals seeking Preliminary guidance about their symptoms and possible conditions before visiting a doctor or pharmacy.



Families & Caregivers

Parents or caregivers looking for basic OTC medicine recommendations and home-care advice for family members.



Students & Healthcare Learners

Medical and pharmacy students who want to explore diseases, symptoms, and OTC medications for educational purposes.



KEY FEATURES (What does PharmacyGPT offer?)



AI Symptoms Analysis

Analyzes symptoms using AI to predict possible medical conditions.



OTC Medicine Recommendations

Suggests suitable non-prescription medicines available in Egypt.



Egyptian Market Support

Recommends medicines commonly available in Egyptian pharmacies.



Precautions & Contraindications

Reduces the risk of incorrect medicine usage.



HOW FEATURES SOLVE USER PROBLEMS



General Users



Get fast, AI-powered insights about their symptoms and receive safe OTC medicine recommendations without the need for immediate doctor visits.



Families & Caregivers



Make informed decisions for their family's health with reliable medicine suggestions, precautions, and home-care guidance.



Students & Healthcare Learners



Learn and explore real-world medical cases, symptoms, and OTC treatments in an interactive and practical way.



Home-Care Recommendations

Provides self-care guidance for minor conditions.



AI Chatbot

Allows users to ask follow-up questions and receive additional guidance.



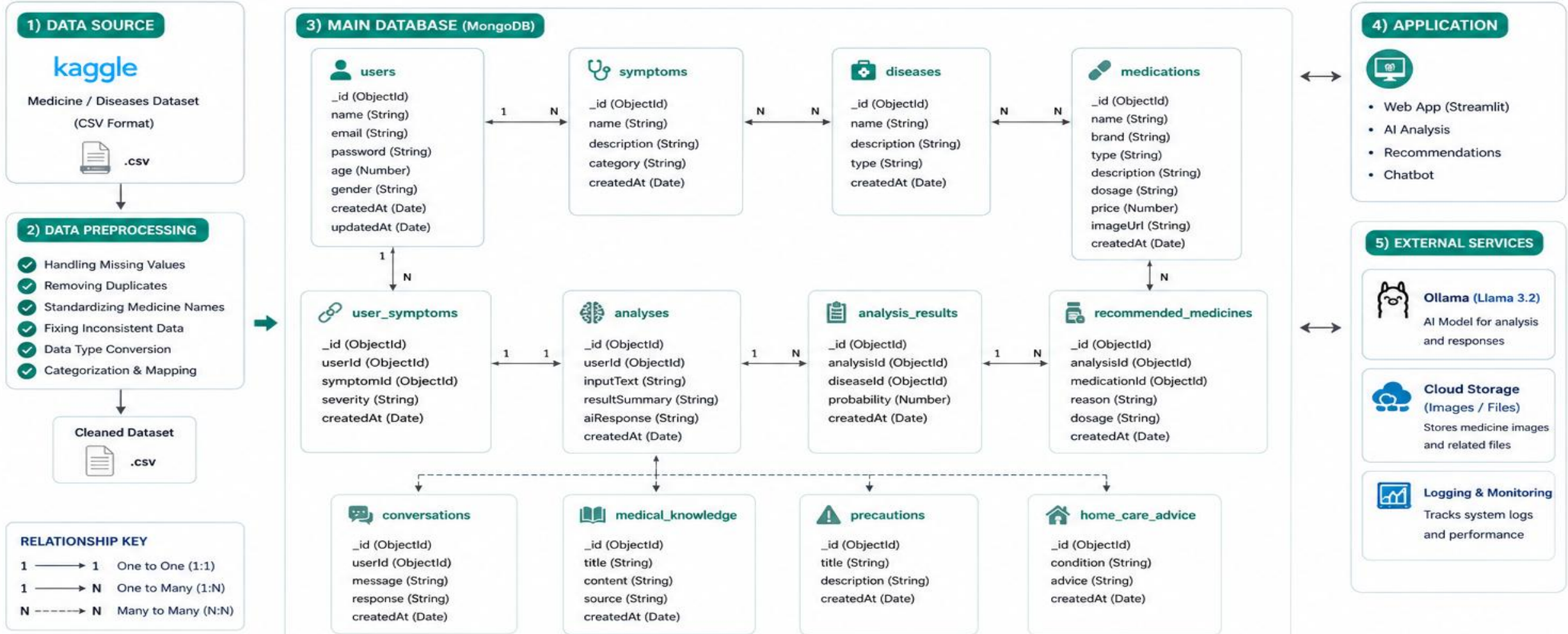
AI Symptoms Analysis

Helps users understand possible conditions based on symptoms.

PHARMACY GPT – DATABASE ARCHITECTURE



Pharmacy GPT uses a cleaned and structured dataset from Kaggle (originally CSV format).
The data is stored in MongoDB and used by the application and AI model for analysis and recommendations.



Programming Languages & Frameworks

The core technologies and tools used to build PharmacyGPT

1) PROGRAMMING LANGUAGES



Python

Backend development
and AI logic



JSON

Medical knowledge
storage format



CSV

Dataset format used
for storing data

2) FRAMEWORKS & LIBRARIES



Streamlit

Web application framework for
building the user interface



Pandas

Data cleaning, processing
and analysis



LangChain

LLM orchestration and
prompt management



Ollama (Llama 3.2:3B)

Runs the local LLM model
for reasoning and responses

3) SUPPORTING TECHNOLOGIES



Streamlit

Web application framework for
building the user interface



CSV Storage

Cleaned and structured data
stored in csv files



Git & GitHub

Version control and
collaboration



VS Code

Development environment
and code editor

Live Application

Programming Languages & Frameworks

The core technologies and tools used to build PharmacyGPT



1) MAIN DELIVERABLES



PharmacyGPT
Web Application



Technical Report



Project Documentation



Cleaned & Structured
Medical Dataset (CSV)



AI Knowledge Base
(Ollama – Llama 3.2)



Source Code
(GitHub Repository)



2) PROJECT MILESTONES



Dataset Collection (Kaggle)
20 Feb 2026



Data Cleaning & Preprocessing
3 Mar 2026



Application Development
10 Mar 2026



AI Integration (Ollama)
20 Mar 2026



Testing & Validation
30 Mar 2026



Final Documentation
25 Jun 2026



Project Delivery
3 Jul 2026



3) FINAL PRODUCTS



Working PharmacyGPT Application
Fully Functional Web App



Project Report
Complete Technical report (PDF)



Complete Source Code
Available on GitHub



Cleaned Medical Dataset
Preprocessed CSV file



Presentation Slides
All project presentation materials

Project Team & Roles



TEAM MEMBERS



Mahmoud Hatem
Team Leader

Planning + AI + Testing



Abdullah Elgondakly
Backend Developer

Python + Processing



Kerolos Nagih
Frontend Developer

Streamlit + UI



Ehab Maher
Dataset & Ollama

Data Engineer



Kareim Abdelomnem
Testing & Reports

Documentation & Testing



COLLABORATION



GitHub

Version control, source code management, and collaborative development.



WhatsApp

Daily communication, quick updates, and task coordination.

Thank You!

Thank you for your time and attention.
We appreciate your interest in our project.

Questions or Feedback?



We would be happy to answer any questions or hear your feedback.

CONTACT US



GitHub

github.com/m-hatem-lgtm/DEPI-Graduation-Project.git